

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (EC)

December 2012

EC 702 Digital Communication & Coding

Date: 18.12.2012, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define Random variable & Probability Distributions. [04]
(b) How do you represent band pass signal? [03]
- Q - 2 (a) Derive Central limit theorem. [06]
(b) Explain correlation demodulator. [05]
(c) Draw only pulse shapes for full response CPM and partial response CPM. [03]

OR

- Q - 2 (a) How do you design signal for band limited channel? [06]
(b) Explain characterization of band limited channel. [05]
(c) What is conditional probability? [03]
- Q - 3 (a) Explain any one memoryless modulation method. [07]
(b) Derive equation for average error probability for binary signals. [07]

OR

- Q - 3 (a) Explain multichannel digital communication in AWGN channel. [07]
(b) Explain multicarrier communication system with block diagram. [07]

SECTION - II

- Q - 4 (a) Find the generator polynomial for $g(x)$ for a (7,4) cyclic code and find code vector [04]
for the following data vector 1110, 0000, 1011, 1100.
- (b) Construct a systematic (7, 4) cyclic code using generator polynomial $g(x) = x^3 + x^2 + 1$, [03]
For data vector $d = 1010$.

- Q - 5 (a) Design the code tree for coder given as $N=3$ & $v=2$ where [07]
 $v_1 = S_1 \oplus S_2 \oplus S_3$, $v_2 = S_1 \oplus S_3$
 Take input sequence as 11010 OR 10101
- (b) Explain decision feedback equalizer with block diagram. [07]

OR

- Q - 5 (a) Derive convergence properties of LMS algorithm. [07]
 (b) Explain binary PSK receiver with block diagram. [07]
- Q - 6 (a) A signal amplitude x is a random variable uniformly distributed in the range $(-1,1)$. [04]
 This signal is passed through an amplifier of gain 2. The output y is also a common variable, uniformly distributed in the range $(-2,2)$. Determine entropies $H(x)$ and $H(y)$
- (b) A zero memory source emits six messages with probabilities 0.3, 0.22, 0.18, 0.15, 0.1 and 0.05. Find the 4-ary Huffman code. Determine its average word length, the efficiency and the redundancy. [05]
- (c) A memoryless source emits messages m_1 and m_2 with probabilities 0.8 and 0.2 respectively. Find the Huffman binary code for this source as well as for its second and third order extensions. Determine the code efficiencies in each case. [05]

OR

- Q - 6 (a) Explain squaring loop with block diagram. [05]
 (b) Explain tomlinson-harashima precoding with block diagram. [05]
 (c) What is liner transversal filter? [04]